

Development Time and AI Assistance Disclosure

Development Time

The project took approximately 20–25 hours in total.

- Understanding the assessment requirements and designing the approach: ~3 hours
- Reviewing and adapting concepts from a previous Vehicle Health Monitoring project: ~2 hours
- Data ingestion and drive-level processing pipeline: ~3 hours
- Cooling-system parameter engineering and health-score design: ~5 hours
- Trend analysis, anomaly detection, and predictive maintenance logic: ~4 hours
- Visualization, storage architecture (CSV + JSON), and reporting: ~3 hours
- Dockerization, testing, debugging, and documentation: ~4 hours

Previous Project Experience

The foundation for this assessment was influenced by my earlier project, **AI-Powered Vehicle Health Monitoring & Predictive Maintenance System**, which focused on real-time OBD-II data collection, anomaly detection, predictive maintenance, and vehicle diagnostics using machine learning and deep learning techniques.

That project used Autoencoders, LSTM networks, and XGBoost models to detect anomalies and predict failures from multivariate vehicle telemetry. The experience gained from that work helped me understand vehicle sensor behavior, thermal-system monitoring, health scoring, anomaly detection, and predictive maintenance concepts.

For this assessment, I did not reuse the previous implementation directly. Instead, I adapted the engineering concepts to the cooling-system-specific requirements and designed a new approach centered around drive-comparable cooling-health parameters, trend analysis, parameter drift detection, SPC monitoring, and predictive maintenance.

Use of AI Assistance

I used AI coding assistance primarily for software engineering support, including:

- Code refactoring and organization
- Docker configuration and troubleshooting
- Plotting and visualization improvements
- Debugging support
- Documentation refinement

I also used the AI coding assistant Windsurf for implementation assistance and productivity improvements.

The engineering decisions—including parameter selection, normalization strategy, health-score formulation, trend-analysis design, anomaly-detection logic, and justification of parameter comparability across drives—were designed, evaluated, and validated by me.

The most significant engineering challenge was determining how to create cooling-system health parameters that remain comparable across drives with different operating conditions and how to detect degradation before a critical threshold is reached. The final design reflects my own reasoning, experimentation, and iterative refinement of the detection logic.